

- Bobby, 25-year-old male comes for STI testing at the after unprotected sex.
 - A complete history and physical examination, including anogenital exam, is unremarkable
 - You recommend tests for HIV, syphilis, gonorrhea, Chlamydia, HBV and trichomonas.
 - Despite unprotected exposures and the USPSTF recommendation that all between the ages of 15 and 65 have at least one HIV test and, he has refused HIV testing previously because he believes HIV is a government plot.
- Where do you tell Bobby HIV came from?

Bobby

- Bobby reported unprotected sex with multiple women and men, has been treated for multiple STIs including gonorrhea and syphilis and has recurrent genital herpes. He shared needles with others when using heroin, last 12 years ago.
- What do you estimate is Bobby's risk of acquiring HIV infection?

Question 9

Match the ART with its mechanism of action

- | | |
|-----------------|---|
| 1. Maraviroc | A. False nucleoside for reverse transcriptase |
| 2. Dolutegravir | B. Inhibits action of gp41 |
| 3. Enfuvirtide | C. Inhibits reverse transcriptase and integrase |
| 4. Tenofovir | D. Prevents viral maturation |
| 5. Rilpivirine | E. Prevents formation of provirus |
| 6. Atazanavir | F. Prevents transcription of viral RNA |
| 7. Emcitribine | G. Binds CXCR4 |
| 8. Biktarvy® | H. Binds CCR5 |

Question 12

In persons living with untreated HIV infection, at what CD4 count does humoral and cell-mediated immunity precipitously fail?

1. 50
2. 100
3. 200
4. 500

Question 13

Symptoms of the acute retroviral syndrome include all of the below except

1. icterus
2. fever
3. adenopathy
4. oral or vaginal candidiasis

Question 14

Persons living with latent HIV infection are at risk for all the following except

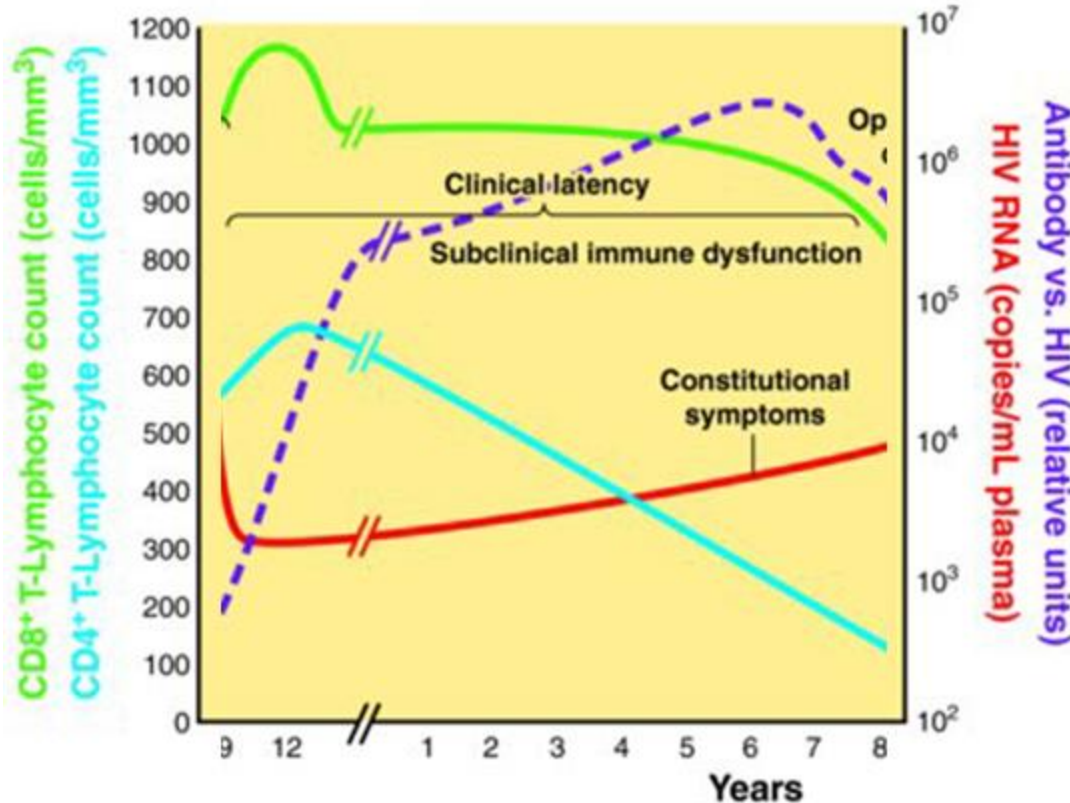
1. premature cardiovascular disease
2. fatigue
3. cytomegalovirus retinitis
4. transmitting HIV to others

Latent HIV infection

Asymptomatic,
transmissible
HIV infection

Progressive risk of
CV disease
Renal disease
Metabolic syndrome
Hyperlipidemia
Fat redistribution
Metabolic acidosis

“Immune lacunae”



1 in 7 persons with HIV are unaware of their infection

Question 15

All of the following are AIDS-defining illness except:

1. invasive cervical cancer
2. HIV-associated wasting syndrome
3. chronic cryptosporidiosis
4. Hodgkin lymphoma

- What HIV test do you offer Bobby?
 - HIV 4th generation antibody-p24 antigen, which is positive. He is understandably upset. You counsel him, recommend 100% barrier protection during sex, and let him know that with behavior modification and treatment he could expect to live to old age.
- What other tests should you order?

Question 16

Persons in whom HIV-1 infection is highly suspected but 4th gen antibody/antigen testing is negative should be tested by

1. Repeat 4th gen ab/ag
2. HIV-1 viral load
3. CD4 count
4. No further testing

Tests in Persons Newly Diagnosed with HIV

- CD4 cell count, HIV viral load
- HIV-genotype, integrase genotype, coreceptor tropism
- STIs: gonorrhea/chlamydia (TMA), trichomonas (PCR), syphilis (RPR)
- Tuberculosis screening (skin test, IGRA)
- Hepatitis A (total antibody),
Hepatitis B (surface antigen, surface antibody, core total ab),
Hepatitis C (antibody)
- CD4 <100: CMV IgG, Toxoplasma IgG

- Bobby's tests show CD4 is 150/ml and HIV viral load is 150,000/ml. HIV-resistance testing is negative (susceptible to all ART). Hepatitis B surface antibody is reactive and all other testing was negative.
- Should Bobby start treatment and if so, why now and with what medications?

Question 17

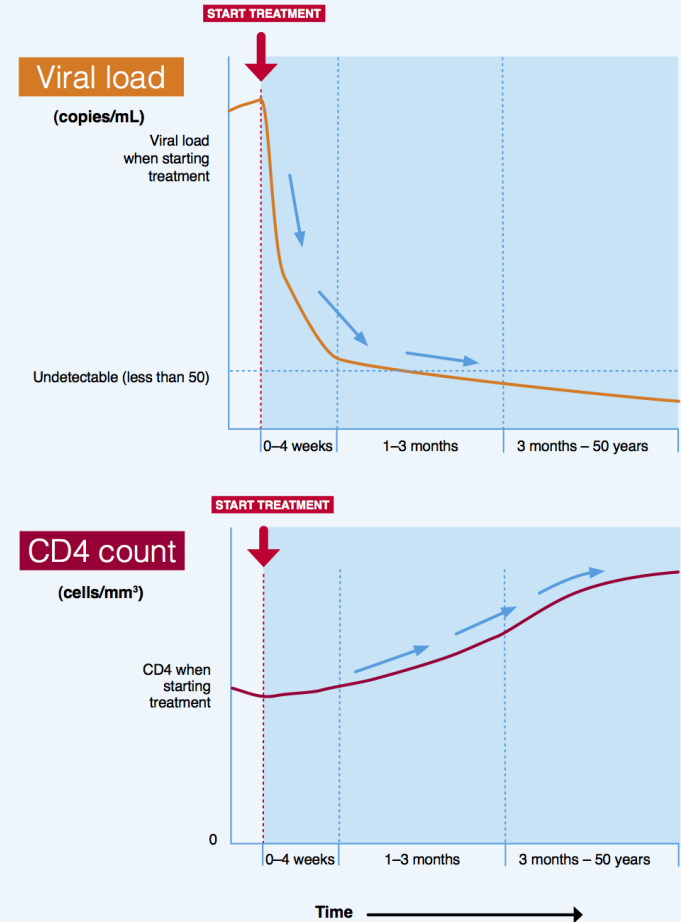
In persons treated for HIV, the viral load is predictive of

1. risk of mortality
2. risk of opportunistic infection
3. risk of malignancy
4. ART resistance

Bobby

- Bobby begins bictegravir, emtricitabine & tenofovir alafenamide (fixed combination) daily
 - Counsel 100% compliance with ART and safe sex
 - TMP/SMZ DS 3x weekly for PJP prophylaxis
 - Pneumococcal, tetanus, meningococcal and HPV vaccinations

HIV after starting ART



Bobby

- Bobby returns 1 month later. He had no side effects and has been compliant (he set an alarm on his phone as a reminder).
 - CD4 is 220/ml and viral load is 1500/ml.
 - You explain that he can expect an undetectable viral load about 6-8 weeks after starting ART.
- He feels great but explains that he hates taking pills everyday, asks if he needs to do this for the rest of his life and will there ever be a cure. What do you tell him?

Bobby

- You tell Bobby that there is no cure, but as long as he is compliant and HIV viral load remains undetectable in the future there is the possibility for once-a-month cabotegravir/rilpivirine.
 - He stops TMP/SMZ after CD4 >200 and viral load is undetectable for 6 months on ART.
- 1 year later, he is in a monogamous, discordant relationship and asks how to prevent HIV infection in his partner. What do you tell him?

HIV: Summary

- HIV originated in Africa due to encroachment of forested land exposed humans to SIV.
- HIV is transmitted via blood, sexual and prenatal exposures, all of which can be prevented.
- The viral life cycle can be interrupted at many points by ART to achieve an undetectable viral load.
- CD4 depletion results in progressive and eventually severe immune suppression increasing the risk of infection and malignancy.
- The HIV pandemic has resulted in >40 million deaths and remains active but there are many reasons to be hopeful that it will end during out lifetime.

Bobby

- Bobby does very well.
- At 5 years CD4 is 775/ml and viral load has remained undetectable.
- He has not missed a single dose of ART.
- His partner is on PrEP and is uninfected.